

Employee Analytics

Employee Attrition Prediction

Using Machine Learning & Data-Driven HR Analytics

BY: Lileshwar Mahajan

Introduction

- Employee attrition is one of the most costly challenges in modern organisations — replacement costs can reach 50%–200% of annual salary.
- With the growing availability of HR data, machine learning enables a structured, data-driven approach to understand patterns and predict turnover.
- This project focuses on building a predictive model for employee attrition using classification algorithms to support proactive retention strategies.
- Implemented a full ML pipeline including data cleaning, exploratory analysis, feature engineering, SMOTE-based class balancing, and model training.
- Four models were evaluated: Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting — with the best selected by ROC-AUC.

Objective

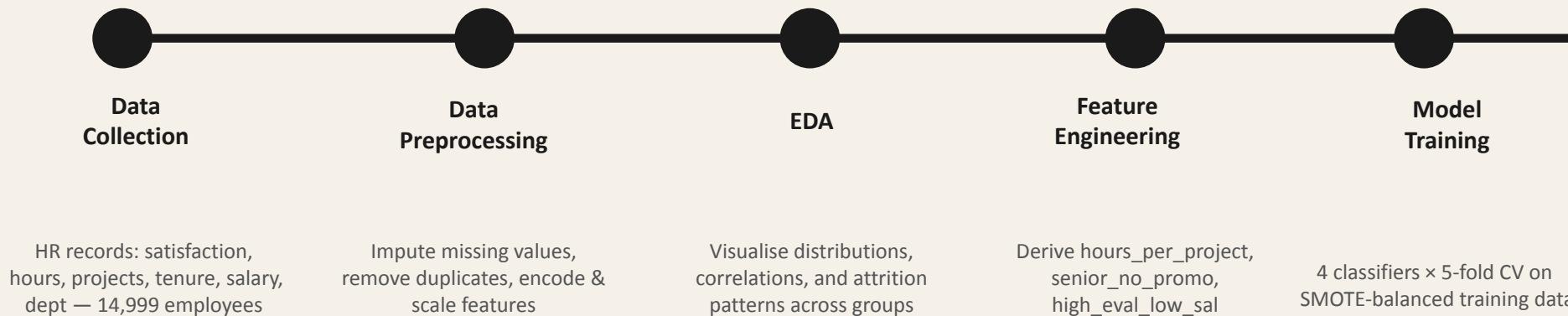
- Develop a data-driven framework to predict which employees are at risk of leaving the organisation.
- Apply and compare multiple classification algorithms to identify the most accurate and reliable model.
- Perform thorough exploratory data analysis to uncover the key drivers of employee attrition.
- Handle class imbalance using SMOTE to ensure fair and unbiased model training.
- Translate model predictions and feature importance scores into actionable business recommendations for HR teams.
- Build a scalable, interpretable pipeline deployable as a monthly attrition risk scoring system.

Motivation

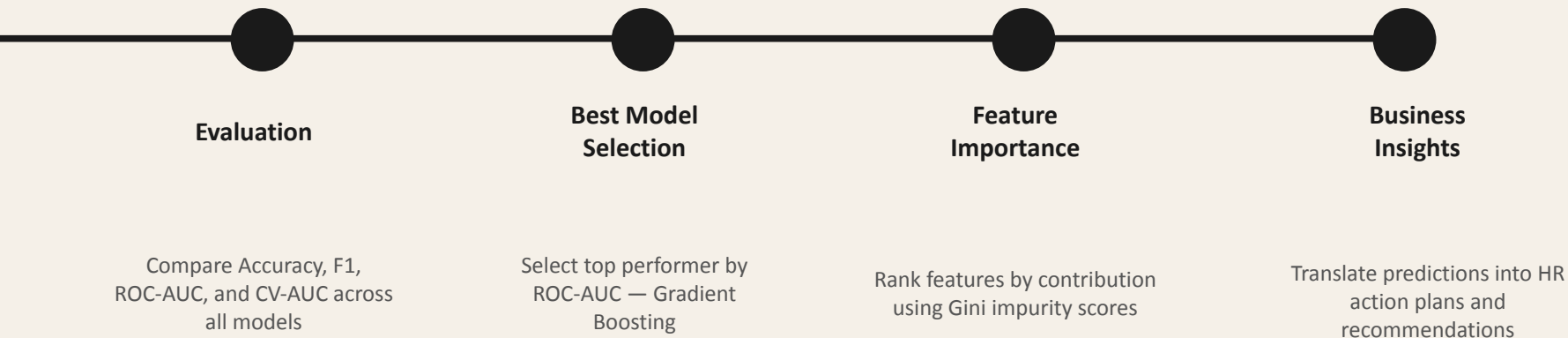
- Traditional HR decisions are largely reactive — organisations act only after an employee resigns, losing valuable talent and institutional knowledge.
- Employee satisfaction, workload, compensation, and career growth are measurable signals that can predict departure before it happens.
- A predictive model shifts the organisation from reactive to proactive, enabling targeted retention efforts where they matter most.
- Indian and global enterprises face significant attrition challenges, particularly in technical and sales departments.
- A robust, interpretable framework is more valuable than a single model — combining multiple signals improves reliability and trust.

"Move from reactive exits to proactive retention — intervene before the resignation, not after."

Approach



Approach



System Architecture

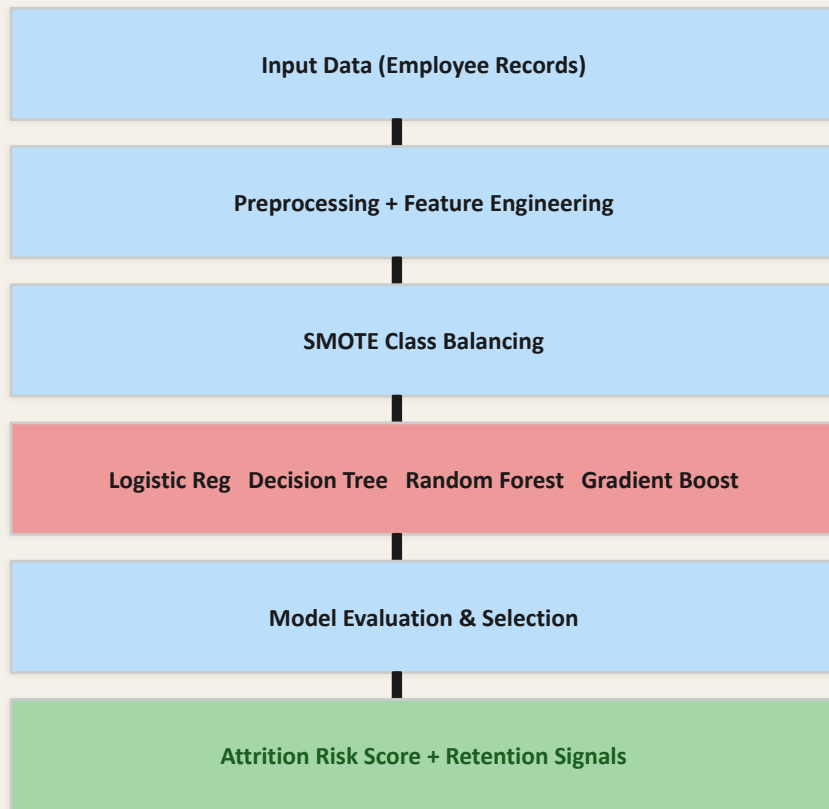
- Input: historical employee records (14,999 rows, 10 features).
- Data cleaning: missing value imputation, duplicate removal.

Feature engineering: 3 derived business-logic features added.

Encoding: ordinal for salary, one-hot for department.

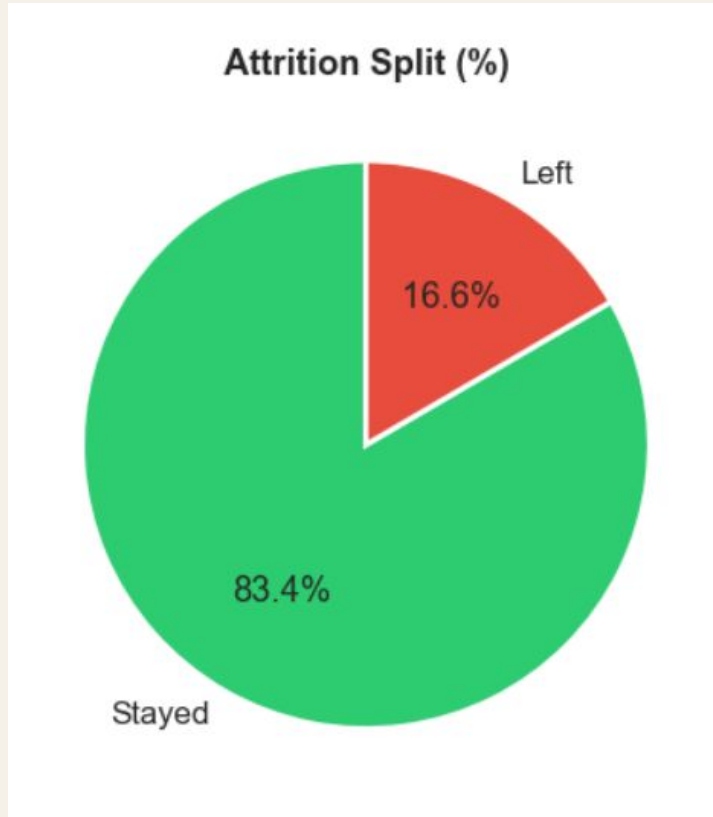
- SMOTE applied to training set only to balance the class split.
- Four classifiers trained with 5-fold cross-validation.
- Best model selected by ROC-AUC and deployed for

Stack ensemble: models capture different HR signals — combined into one risk score.



Exploratory Data Analysis

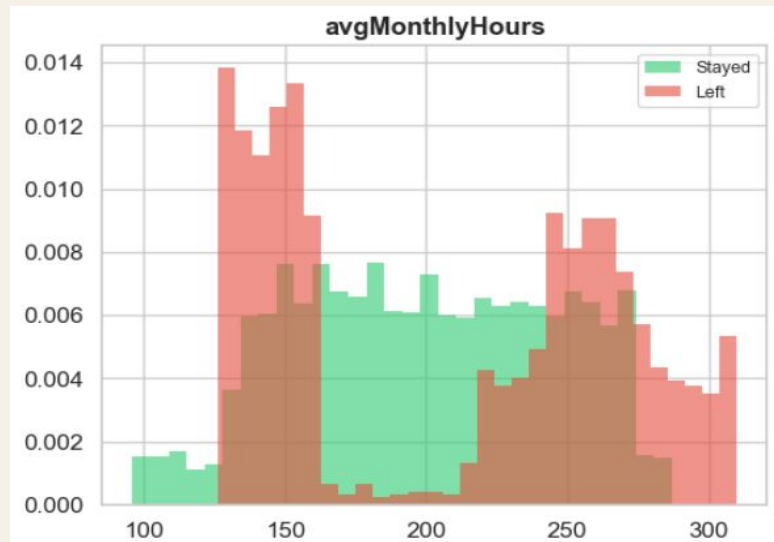
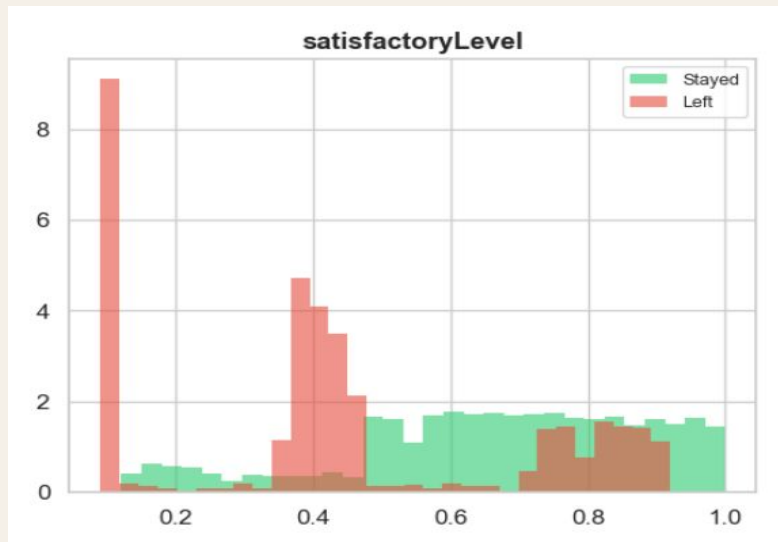
Target Variable: Employee Attrition



- 14,999 employee records across 10 departments.
- 16.6% of employees left — a significant but manageable attrition rate.
- Class imbalance (83.4% Stayed vs 16.6% Left) present — requires SMOTE correction before training.
- Without balancing, most models would default to predicting 'Stayed' for all employees, missing at-risk individuals entirely.

"Class imbalance is the first challenge — SMOTE generates synthetic minority samples on training data only."

Feature Distributions: Stayed vs Left



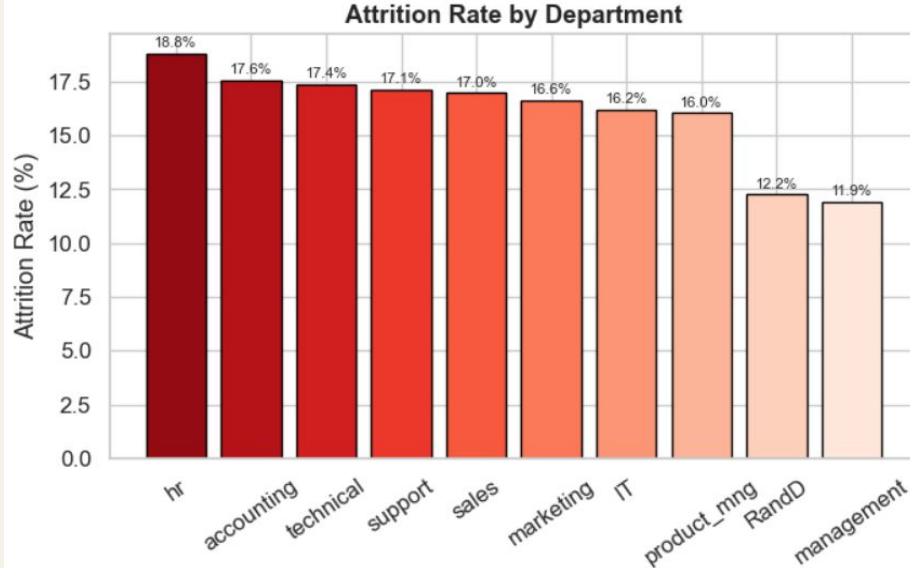
Satisfaction level: employees who left cluster sharply below 0.40 — the single strongest predictor.

Monthly hours: two attrition clusters appear — moderate overwork AND extreme burnout zones.

- Features with large separation between the two distributions carry the most predictive power.

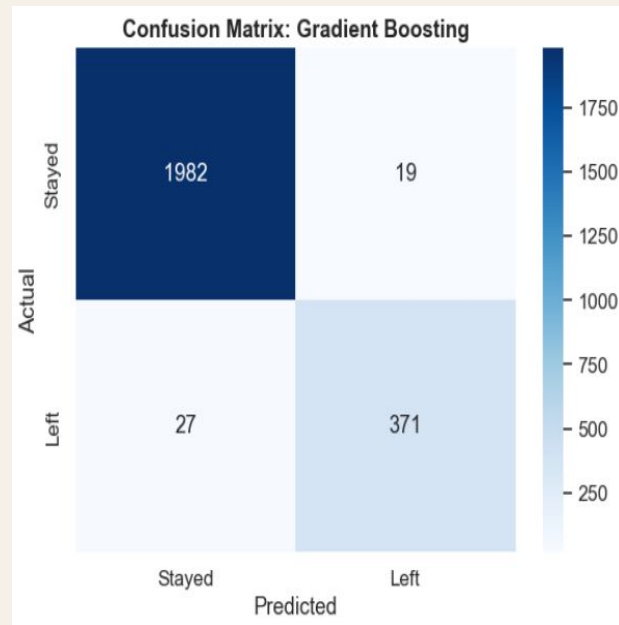
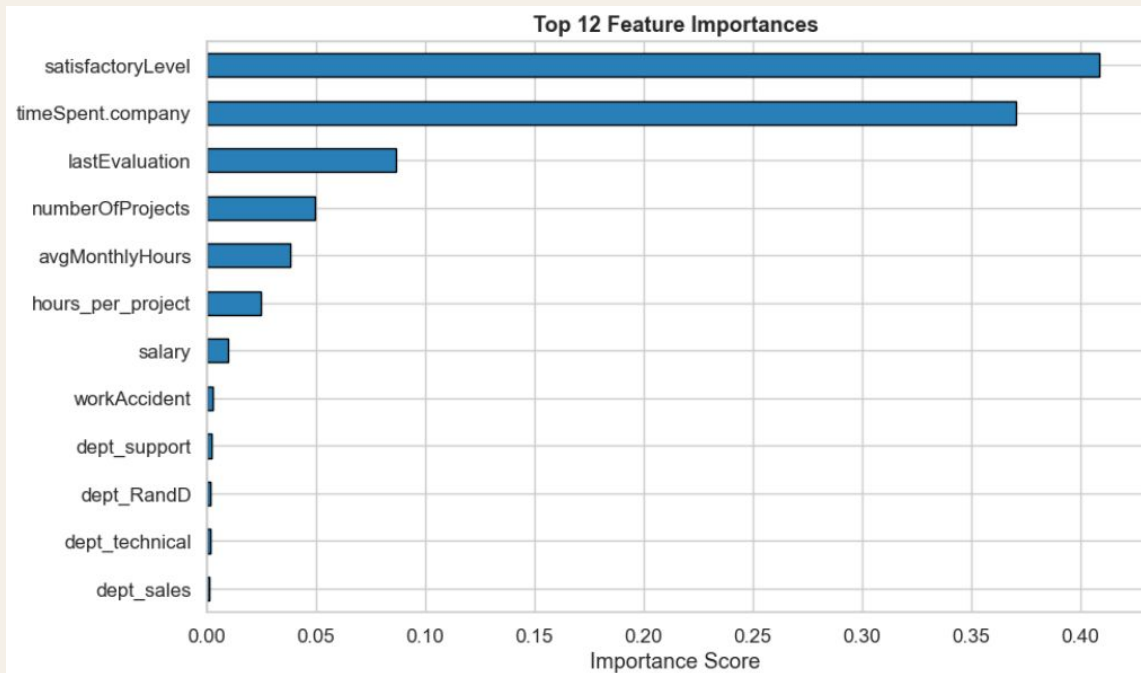
Attrition by Department and Salary

Attrition Rates Across Categorical Features

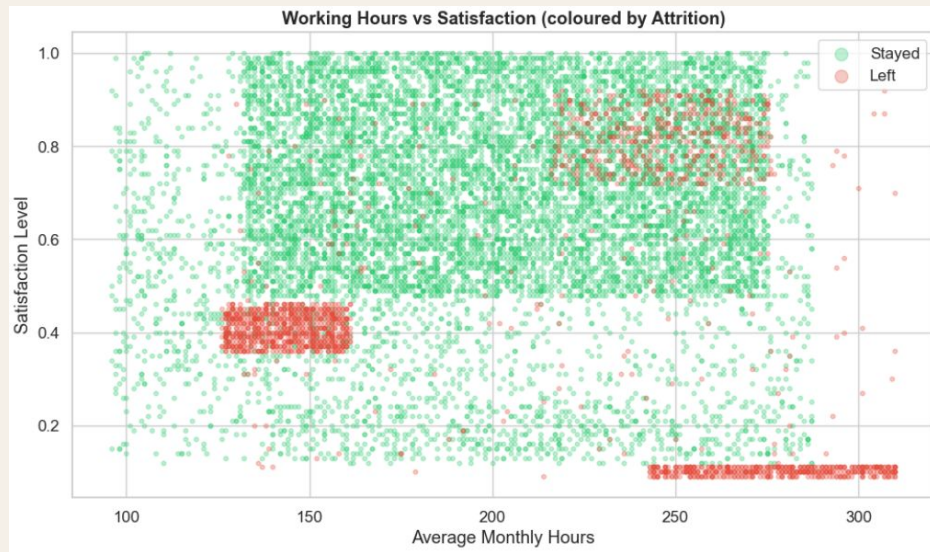
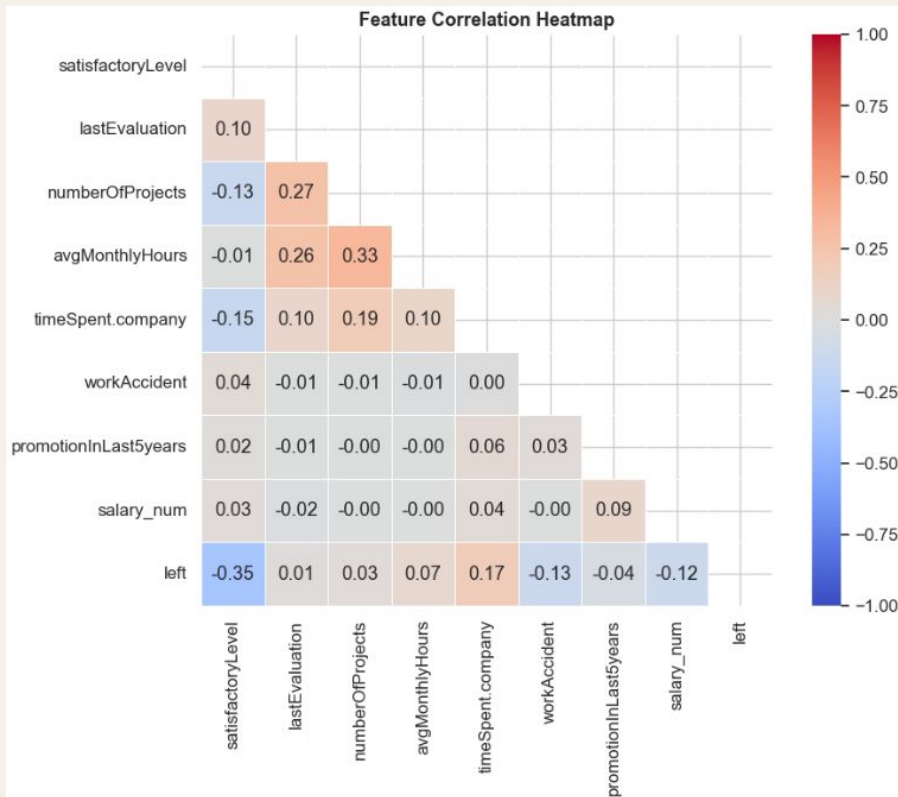


- Low-salary employees leave at significantly higher rates than medium or high-salary peers.
- Certain departments (e.g., Sales, Technical) show elevated attrition and need targeted programmes.
- Salary level is an ordinal feature — encoded as low=0, medium=1, high=2 to preserve natural order.

EDA Process Output Graphs



Correlations and Cluster Patterns



Heatmap: satisfaction shows the strongest negative correlation with attrition.

Scatter: two distinct left-clusters visible — dissatisfied employees and overworked employees.

Feature Engineering

Three new features were derived from raw columns, each grounded in a clear business logic.

hours_per_project

`avgMonthlyHours /
numberOfProjects`

High value = each project demands too much time. Signals workload overload.

senior_no_promo

`timeSpent > 5 AND
promotion = 0`

Senior employee with no recognition. Indicates career stagnation and disengagement.

high_eval_low_sal

`lastEvaluation > 0.75 AND
salary = low`

High performer on low pay — the strongest flight risk signal in the dataset.

Model Performance & Comparative Analysis

Models Used

Logistic Regression

Baseline / Linear

Simple, interpretable linear baseline. Estimates attrition probability as a logistic function of features. Sets the performance floor.

Decision Tree

Baseline / Non-linear

Rule-based model using recursive feature splits. Easy to explain to non-technical stakeholders. Max depth = 8.

Random Forest

Bagging Ensemble

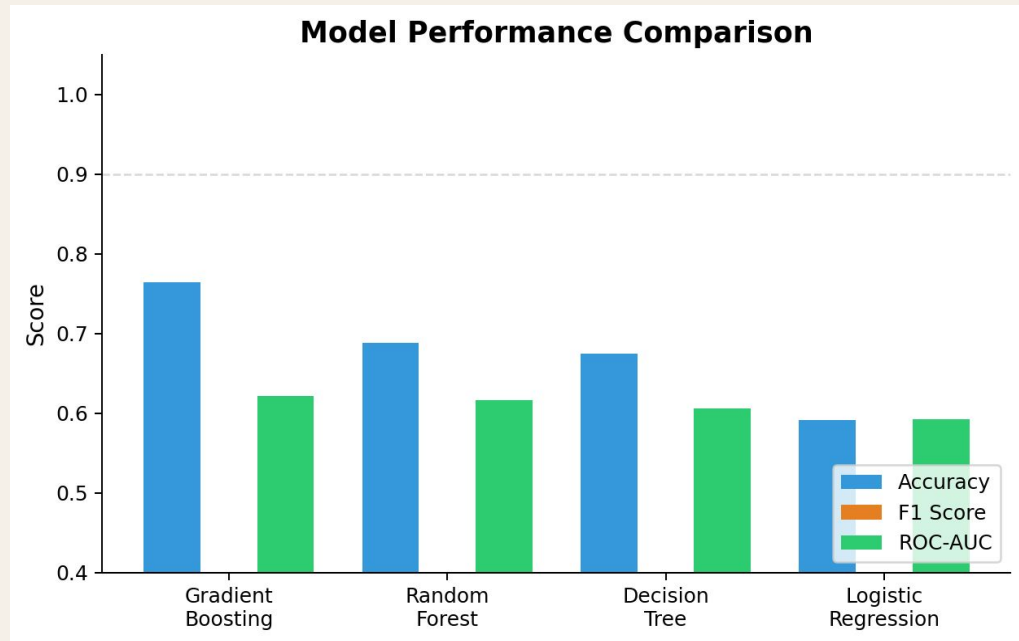
200 trees trained on random data and feature subsets. Averaging many trees reduces overfitting and improves stability.

Gradient Boosting

Boosting Ensemble

200 trees built sequentially — each correcting prior errors. Learning rate = 0.05, subsample = 0.8. Best overall performer.

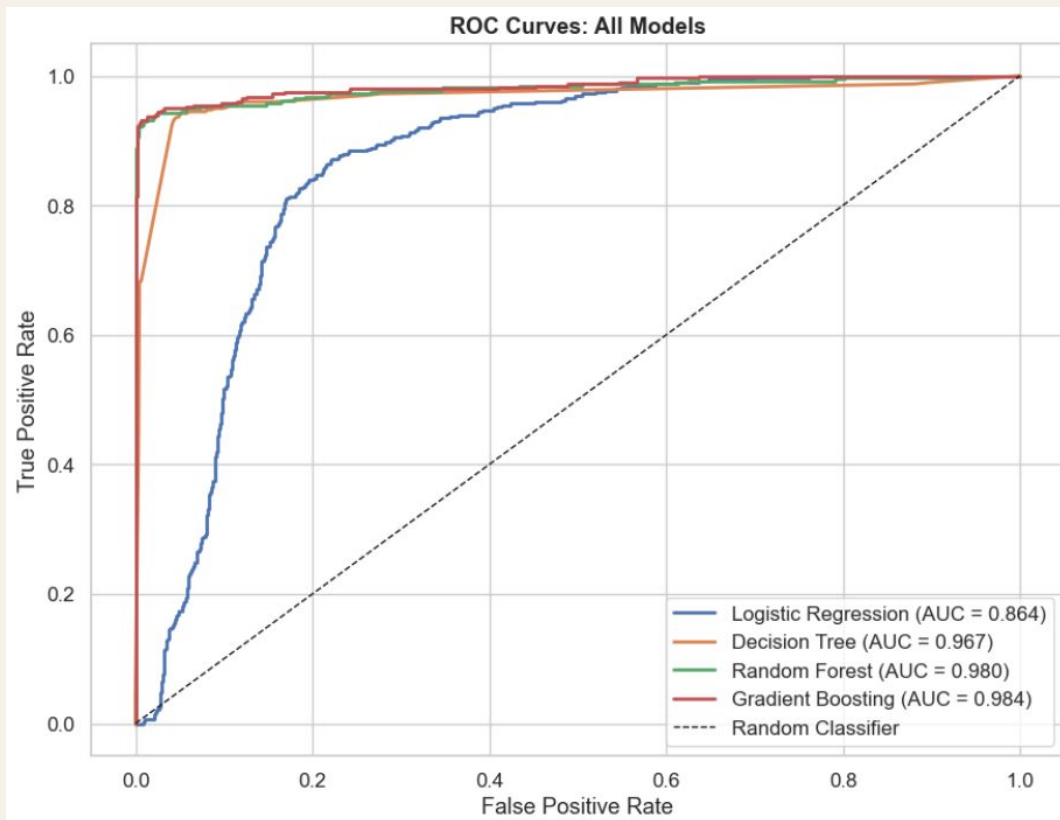
Model Performance Comparison



	Model	Accuracy	F1 Score	ROC-AUC	CV-AUC (train)
1	Gradient Boosting	0.9808	0.9416	0.9845	0.9970
2	Random Forest	0.9825	0.9459	0.9805	0.9967
3	Decision Tree	0.9416	0.8430	0.9668	0.9819
4	Logistic Regression	0.8003	0.5846	0.8635	0.8632

- Best model: Gradient Boosting (AUC = 0.9808)
- Boosting outperforms all baselines across accuracy and AUC metrics.
- High CV-AUC vs test AUC gap in Random Forest suggests mild overfitting — Gradient Boosting is more stable.

ROC Curves: All Models



- A curve closer to the top-left corner = better model. Diagonal line = random guessing.
- All four models outperform random, with Gradient Boosting achieving the highest AUC.
- The gap between models is modest — ensemble approaches would close this further.
- AUC measures model ranking ability across all thresholds, not just at a fixed cut-off.
- The best threshold for deployment depends on business cost: missing a leaver (FN) vs false alarm (FP).

Practical Implications

- Provides a data-driven attrition risk score for every active employee on a monthly basis.
- Enables HR teams to focus retention budgets on the highest-risk individuals rather than guessing.
- Satisfaction surveys, workload monitoring, and compensation reviews become targeted — not blanket.
- The trained model can be serialised (pickle / joblib) and integrated into existing HRIS platforms.
- Scalable to different business units, geographies, or companies with minimal re-training.
- Outputs are interpretable — feature importance allows managers to understand why an employee is flagged as high risk.
- Can be extended to portfolio-level workforce planning and proactive succession planning.

"Bridges the gap between theoretical HR models and real-world people decision-making."

Key Risk Factors & Recommendations

Key Risk Factors

Low satisfaction (<0.40): strongest single predictor.

Overwork (>250 hrs/month): burnout signal.

High performer + low salary: flight risk.

Senior + no promotion: disengagement.

6+ concurrent projects: overload indicator.

Recommendations

- Quarterly satisfaction surveys with mandatory follow-up plans.
- Soft cap of 4–5 concurrent projects per employee.
- Salary review for high-performers in the low band.
- Structured promotion pathways for employees past year 4.
- Flag >250 hrs/month for automatic manager check-ins.
- Monthly attrition risk scores surfaced to HR business partners.

Limitations & Future Scope

Limitations

- Relies on historical data and may not capture sudden org changes or market shocks.
- External factors (economic conditions, competitor offers) are not included.
- Dataset is synthetic — real HR data would yield richer and more reliable signals.
- Model does not account for the cost asymmetry of False Negatives vs False Positives.
- No temporal modelling — satisfaction trajectory over time is not captured.

Future Scope

- Integrate real-time HRIS data for dynamic monthly risk updates.
- Incorporate sentiment analysis from internal surveys and communication tools.
- Apply SHAP for per-employee, locally interpretable risk explanations.
- Use LSTM or Transformer models to capture satisfaction trend over time.
- Extend to portfolio-level workforce planning and automated retention workflows.

Conclusion

- Developed a complete, end-to-end ML pipeline for predicting employee attrition from structured HR data.
- Statistical EDA revealed that satisfaction level, working hours, and salary band are the dominant attrition drivers.
- Four classification models were trained and evaluated — Gradient Boosting achieved the best ROC-AUC of 0.9808.
- SMOTE balanced the class split without contaminating the test set, ensuring realistic evaluation.
- Feature engineering added measurable predictive value — all three derived features appear in the top-12 importance rankings.
- Model outputs can be deployed as a monthly risk-scoring pipeline, surfacing the top at-risk employees to HR teams.

"This approach bridges the gap between traditional HR intuition and modern data-driven people analytics."

Thank You

People Analytics: Employee Attrition Prediction

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